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Remanent magnetization of Precambrian and Cretaceous kimberlites in South Africa

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Summary. Middle Precambrian and Cretaceous kimberlites were collected from three sites (Premier, Montrose and National) and two sites (Wesselton and Koffyfontein) in South Africa respectively. The natural remanent magnetization of these rocks remains stable to both alternating field and thermal demagnetization. The virtual geomagnetic pole-positions derived from the directions of stable remanence of the Precambrian rocks can be correlated with palaeomagnetic poles obtained from other Middle–Late Precambrian rocks in Africa. The Cretaceous poles for the Wesselton and the Koffyfontein rocks coincide with other Cretaceous poles.

1 Introduction

Palaeomagnetic surveys in Africa covered a range of geological time from the middle Precambrian to the Quaternary (McElhinny *et al.* 1968; Creer 1970), and include a few determinations from kimberlites. The formation of kimberlites is characterized by an origin deep in the upper mantle and a rapid rise through the crust with fast cooling near the surface. Accordingly their remanent magnetization should have been acquired within a short time and have properties similar to the thermoremanent magnetization of normal volcanic rocks.

In 1973 September, the International Conference on Kimberlite was held in Cape Town. One of us (KS) accompanied the field excursion to collect oriented samples of kimberlites from several diamond mines in South Africa. This report describes results on the measurements of natural remanent magnetization (NRM) of these rocks.

2 Collecting sites

Middle Precambrian kimberlites were collected at the Premier Mine (25° 42′ S, 28° 32′ E), the Montrose Pipe (25° 46′ S, 28° 33′ E) and the National Pipe (25° 48′ S, 28° 33′ E).

Cretaceous kimberlites were sampled at the Wesselton Mine (28° 46′ S, 24° 51′ E) and the Koffyfontein Mine (29° 26′ S, 25° 00′ E). The local geomagnetic field around the sampling sites was approximately $D = 343°$, $I = -63°$ and $F = 32000\,\gamma$ (Hermanus Magnetic Observatory 1961).

3 Geology

Five kimberlite sites of different age, mode of emplacement, and type and degree of crustal contamination were selected (Fig. 1).

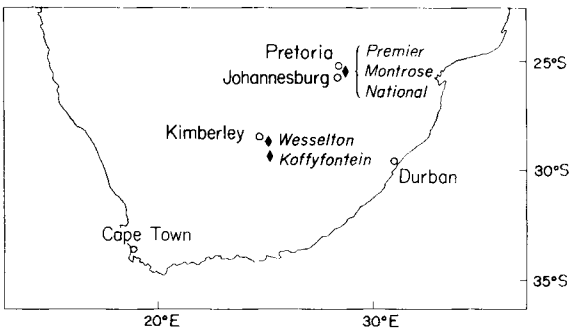


Figure 1. Collecting sites of kimberlites.

3.1 PREMIER MINE, 38 KM EAST OF PRETORIA

The kimberlite pipe of the Premier Mine contains inclusions of Waterberg quartzite and is cut by a thick sill of post-Waterberg diabase (dated at 1115 Myr). Earlier radiometric age is 1750 ± 10 Myr or a minimum 1115 ± 15 Myr (Allsopp, Burger & van Zyl 1967). Recent Rb–Sr age determinations by Barrett & Allsopp (1973) indicate that grey kimberlite is 1250 Myr and brown kimberlite may be as old as 1400 Myr. The pipe measures 860 × 400 m at the surface, decreasing to 820 × 330 m at the 538 m level, with an average dip of 85°. The kimberlites were intruded in at least three distinct phases corresponding to the brown, grey and black varieties. Two samples of grey kimberlite on the 500 m level were collected for this study.

3.2 MONTROSE PIPE, 7 KM SOUTH OF THE PREMIER MINE

The Montrose kimberlite pipe is about 95 m in diameter and consists of a barren core and a diamondiferous rim. The kimberlite was intruded into quartzite and shales of the Transvaal System and a diabase sill of post-Transvaal age. It consists of a dark grey kimberlite breccia containing very few large inclusions. One sample of this kimberlite breccia, on the surface level, was collected.

3.3 NATIONAL PIPE, 11 KM SOUTH OF THE PREMIER MINE

The National kimberlite pipe is 70 m in diameter and was intruded into the quartzites of the Transvaal System and a diabase sill of post-Transvaal age. The pipe contains the following varieties of unweathered kimberlite: (a) a dark basaltic kimberlite with a glassy matrix,

(b) hard, dark grey, basaltic kimberlite, (c) two types of kimberlite breccia with different sizes of inclusions. One sample of variety (a) was collected from the surface level.

3.4 WESSELTON MINE, 10 KM SEE OF KIMBERLEY

Wesselton is one of the five major kimberlite pipes in the immediate vicinity of Kimberley. The kimberlites intrude rocks of the Karroo System, which overlies lavas of the Ventersdorp System with intervening quartzite horizons. According to Allsopp & Barrett (1975), the radiometric age of the Wesselton Pipe is 84 ± 3 Myr.

The surface area of the pipe is $10 \times 10^4\,m^2$. The rocks occupying the pipe can be divided into three groups based on textural and petrographical characteristics. Group 1 contains all non-fragmental varieties of kimberlite, Group 2 is intrusive breccias containing variable amounts of xenolithic material and Group 3 is volcanic breccias having an agglomeratic or tuffaceous appearance. Two samples of Groups 1 and 3 kimberlites were collected from the 600 m level.

3.5 KOFFYFONTEIN MINE, 80 KM SOUTH OF KIMBERLEY

The location of the Koffyfontein, Ebenhaezer and Klipfontein kimberlites along a NW–SE line suggests that their emplacement was controlled by a linear zone of weakness trending in this direction. Both Wagner (1914) and Williams (1932) have suggested that the three occurrences are linked, below surface, by a kimberlite dyke. The kimberlites intrude rocks of the Karroo System, which directly overlie Precambrian granite gneiss.

The Koffyfontein Pipe is roughly circular in shape, having a diameter of about 365 m. In the wall rock adjacent to the pipe, there exists a number of narrow kimberlite dykes. One sample of a kimberlite dyke was collected.

4 Samples

The petrography is given in the Appendix. The weight of each sample was about 800 g. Four to nine cylindrical specimens were drilled from each sample.

5 Magnetic measurements

The NRMs were measured with an astatic magnetometer. The accuracy of measurement was high, but errors in the orientation could be as much as 3°, due to the difficulties of orientating samples in mines. After the measurements of NRM, the specimens were subjected to alternating field demagnetization. The results are shown in Table 1. The NRM was stable

Table 1. Site mean results and apparent poles (north pole).

Locality	Age	NRM		RM after AC demagnetization in 200 Oe				Magnetization of NRM	Pole
		D	I	D	I	K	θ_{95}		
Premier	Middle Pc 1250 Myr	170°	0°	165°	−3°	453	2.7	8.3×10^{-1} mA	169° W 59° S
Montrose	Middle Pc	194	11	193	11	373	2.7	8.2	118° W 66° S
National	Middle Pc	180	−57	185	−58	1410	2.0	130	147° W 26° S
Wesselton	Cretaceous 84 ± 3 Myr	340	−75	340	−71	4240	2.3	2.9 ~ 250	132° W 60° N
Koffyfontein	Cretaceous	348	−66	347	−63	899	1.7	13.5	125° W 71° N

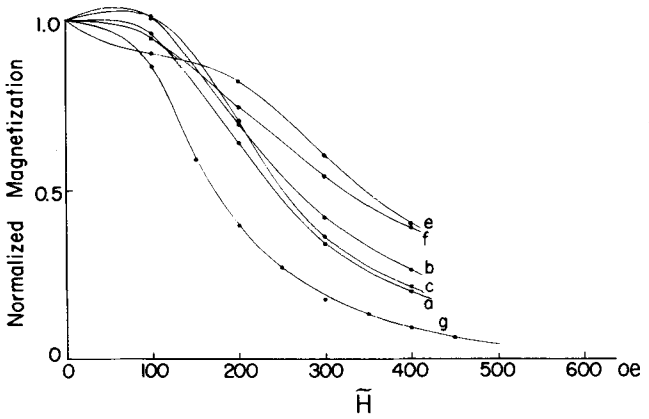


Figure 2. Decrease of intensities of NRM by the application of AC field: (a) grey kimberlite from the Premier Mine (KS-73091705), (b) grey kimberlite from the Premier Mine (KS-73091712), (c) kimberlite from the Montrose Pipe (KS-73091701), (d) kimberlite from the National Pipe (KS-73091703), (e) grey-black kimberlite from the Wesselton Mine (KS-73091902), (f) tuffaceous kimberlite from the Wesselton Mine (KS-73091903), (g) black kimberlite from the Koffyfontein Mine (KS-73092202).

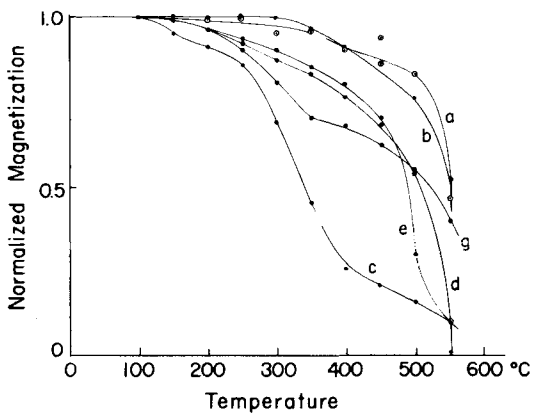


Figure 3. Decrease of intensities of NRM on heating to 550°C. Suffixal letters a, b, c . . . are same as those of subheadings in Fig. 2.

and the directions showed little change, less than 5° in a peak field of 200 Oe and the intensities decreased by approximately 30 per cent of the initial values in the same field (Fig. 2). Fig. 3 indicates the thermal demagnetization of representative samples from each rock type. It is seen that the magnetizations were also stable on heating up to 300°C. This behaviour is often observed in volcanic rocks which retain their primary thermoremanent magnetization since their formation. The thermomagnetic analysis showed irreversible change on heating above 300°C in some rocks. This can be explained by a chemical change of minerals such as oxidization.

6 Discussion

Remanent magnetization is stable to both AC and thermal demagnetization. McFadden & Jones (1977) have noted that some part of the NRM in this kimberlite is attributed to

chemical remanent magnetization (CRM) due to magnetic minerals formed by secondary serpentinization of olivine. The microscopic observations imply that, although a slight influence of CRM is also seen in the present investigation, predominant contributions to NRM are made by either primary or secondary magnetite, both of which crystallized at fairly high temperatures, presumably above its Curie temperature. Accordingly the NRM is apparently largely a TRM, as suggested by Jones (1968).

6.1 MIDDLE PRECAMBRIAN KIMBERLITES FROM PREMIER MINE, MONTROSE AND NATIONAL PIPES

The results show that the magnetization of kimberlites from the Premier Mine is directed to (165°, −3°). A 70–80 m thick post-Waterberg diabase sill cuts across both the pipe and the wall rock, between mainly 380 and 450 m levels, showing an overall dip of 15° to the northeast. The effect of the sill has been to metamorphose the kimberlites both above and below and this effect is petrographically apparent for a distance of some 25 m. Barrett & Allsopp (1973) examined Rb–Sr ages of rocks located at various distances from the contact. They conclude that the heating effect of the sill has to reset the age of micas situated close to the sill and is expected to affect micas farther from the sill to a lesser extent. The influence of the intrusion is geochronologically observable to a distance of 200 m. The distance between the lower contact of the sill and the 500 m level of the pipe is about 50 m and the samples used here are considered to be thermally affected by the intrusion of the sill.

Jones (1968) reported that the stable NRM of kimberlite collected at the 890 ft level of the Premier Mine is directed to (186°, −24°) and that of the post-Waterberg diabase sill to (183°, −3°). The present result on grey kimberlite is (165°, −3°) which is closer to that of the post-Waterberg sill. This trend of the NRM is explained by the secondary heating of kimberlite at the time of intrusion of the sill (Barrett & Allsopp 1973).

The direction of the Montrose Pipe (193°, 11°) is fairly near to that of the Premier but a significant difference is observed between the National (185°, −58°) and the Montrose pipes. The apparent pole positions for these pipes are correlated to either (1) that for the Waterberg succession (Jones & McElhinny 1967) or (2) that for the African igneous rocks *c.* 1300–1000 Myr (Piper 1975, 1976). In case (1), the pole positions for the Montrose and the National Pipes coincide with the upper and the lower horizon of the Waterberg succession. In case (2), the apparent polar wander path for the igneous rocks is close to that obtained by connecting the poles for the National, the Premier and the Montrose in order. In either correlation, it is seen that the Montrose kimberlite can be younger than the Premier, while the National kimberlite is older than the Premier.

Recent contribution by the South African Geological Survey has shown that an extrusive rock, which is part of early Waterberg volcanism, is dated at 1790 Myr and the minimum age of the Waterberg succession is 1420 Myr (Hunter 1974). If the poles for the kimberlites are correlated with the Waterberg, the difference in time of formation between the National and the Montrose Pipes could be several hundred million years. However, the outcrops of these three kimberlites are located within 10 km of one another and the geology of the district shows that these pipes are closely related, hence the difference in time of intrusion is probably small. Although there has been no age determination of the Montrose and the National Pipes, the Premier is well investigated and the rock examined here is dated to be 1250 Myr (Barrett & Allsopp 1973). Thus all the pole positions obtained in this study probably correlate with those of the Middle–Late Precambrian between 1300 and 1000 Myr.

6.2 CRETACEOUS KIMBERLITES FROM WESSELTON AND KOFFYFONTEIN MINES

The new result is close to those of McFadden & Jones (1977) in the case of Wesselton. However, a scatter is observed in two results from the Koffyfontein Mine. McFadden & Jones (1977) have suggested that a movement of rock mass occurred some time during mining operations and this partly contributed to the scatter in their results. However, since we have only one sample we merely note that the virtual geomagnetic pole position for the Koffyfontein is close to that of the Cretaceous mean positions summarized by Creer (1970) and McElhinny & Brock (1975).

6.3 RELATION BETWEEN THE INTENSITY OF MAGNETIZATION AND PETROGRAPHICAL CHARACTER

The intensities of magnetization of dark basaltic kimberlite from the National Pipe and dark grey-black kimberlite from the Wesselton Mine are very strong. As noted in the Appendix, magnetite is one of the main consituents of these rocks and is scattered throughout the groundmass, while it occurs only as an accessory mineral in the groundmass of the other five kimberlites. Hence, the intensities of magnetization correlate approximately with the amount of magnetite contained in the rocks.

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Appendix: petrography of samples examined

A.1 (1) GREY KIMBERLITE (KS-73091705) FROM THE SOUTHEASTERN PART ON THE 500 M LEVEL OF THE PREMIER MINE

Megacrysts and phenocrysts include forsterite and clinopyroxene. Forsterite has mostly been replaced by serpentine and talc. Clinopyroxene has been serpentinized to varying degrees but the majority exhibits cores of unaltered clinopyroxene. Serpentinization has been accompanied by the exsolution of magnetite. The groundmass consists essentially of serpentine and talc with a lesser amount of rutile and phlogopite. Ilmenite, magnetite, chalcopyrite and pyrite are accessory minerals scattered throughout the groundmass.

Rock fragments of biotite-clinopyroxene-serpentine assemblage showing hypabyssal texture with fine-grained phlogopite-clinopyroxene assemblages. Composite fragment of ilmenite-serpentine-rutile assemblage is also present.

A.1 (2) GREY KIMBERLITE (KS-73091712) FROM THE NORTHWESTERN CORNER ON THE 500 M LEVEL OF THE PREMIER MINE

Megacrysts and phenocrysts include orthopyroxene, clinopyroxene, ilmenite, magnetite, phlogopite and perovskite. Orthopyroxene and clinopyroxene have been replaced to varying degrees by serpentine, talc, tremolite and calcite. The groundmass consists mainly of serpentine and tremolite with lesser amounts of ilmenite and magnetite. Pyrite, chalcopyrite and pentlandite occur as fine-grained accessory minerals.

Rock fragments of biotite-ilmenite-magnetite-clinopyroxene assemblage are also found.

A.2 DARK GREY KIMBERLITE (KS-73091701) FROM THE SURFACE LEVEL OF THE MONTROSE PIPE

Megacrysts and phenocrysts include pyrope, pyroxene, phlogopite and forsterite. Pyrope is rimmed by kelyphite. Pyroxene and forsterite have been replaced to varying degrees by serpentine and tremolite with lesser amounts of ilmenite and rutile. The groundmass mainly consists of tremolite and serpentine. Ilmenite, magnetite and haematite are accessory minerals scattered throughout the groundmass.

A.3 DARK BASALTIC KIMBERLITE (KS-73091703) FROM THE SURFACE LEVEL OF THE NATIONAL PIPE

Megacrysts and phenocrysts include orthopyroxene, forsterite, pyrope, clinopyroxene and ilmenite. Orthopyroxene, forsterite and clinopyroxene have been replaced to varying degrees by serpentine and calcite. Pyrope is rimmed by kelyphite. Ilmenite is often bordered by perovskite. The groundmass consists mainly of serpentine, calcite and magnetite. Ilmenite, perovskite and pyrite are scattered throughout the groundmass.

A.4 (1) DARK GREY-BLACK KIMBERLITE (KS-73091902) FROM THE 600 M LEVEL OF THE WESSELTON MINE

Megacrysts and phenocrysts include orthopyroxene, forsterite, white mica, phlogopite and ilmenite. Orthopyroxene and forsterite have been serpentinized to varying degrees. The groundmass consists mainly of serpentine, calcite and magnetite with lesser amounts of perovskite and ilmenite. Magnetite occurs locally as disseminated grains in the groundmass.

Rock fragments of clinopyroxene-orthopyroxene-chlorite-calcite-magnetite assemblages, of calcite-magnetite-chlorite-serpentine-apatite assemblages and of clinopyroxene-serpentine assemblages are also found.

A.4 (2) TUFFACEOUS KIMBERLITE (KS-73091903) FROM THE 600 M LEVEL OF THE WESSELTON MINE

Abundant subangular to subrounded grains ranging from 0.5 to 5 mm long including both altered minerals and individual unaltered minerals. The former include serpentine, chlorite and calcite. The latter include phlogopite, clinopyroxene, orthopyroxene, forsterite, pyrope and ilmenite. Perovskite and magnetite, with small amounts of chalcopyrite, coveline and digenite are present in the groundmass.

A.5 BLACK KIMBERLITE DYKE (KS-73092202) FROM THE OPEN PIT OF THE KOFFYFONTEIN MINE

Large, rounded, anhedral forsterite xenocrysts are clearly distinguishable from much smaller euhedral forsterite microphenocrysts which have mostly been completely serpentinized. The xenocrysts have been serpentinized to varying degrees but the majority exhibit cores of unaltered forsterite. Serpentinization of both forsterites has been accompanied by the exsolution of chromite and magnetite. The groundmass consists essentially of serpentine, phlogopite, calcite, chromite, magnetite, perovskite and zeolite-like mineral. Perovskite, chromite and magnetite are common accessory minerals scattered throughout the groundmass.